CIE 2 – Activity 2 Report ## Simple House Price Prediction using Linear Regression

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Class/Division: T.Y. B.Tech AI & DS / _____
Course: Machine Learning [ADPC509]
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1. Aim To build a Linear Regression model for predicting a continuous house-value target and evaluate it using regression metrics.

2. Dataset The California Housing dataset from scikit-learn (originally StatLib) is used. It contains 20,640 observations, eight numerical predictive features and a continuous median-house-value target. Each observation represents a California census block group from the 1990 census; it is not an individual property listing.

Source: https://scikit-learn.org/stable/datasets/real_world.html#california-housing-dataset

3. Input and target - Input feature (X): `MedInc` – median income in a block group. - Target (y): `MedHouseVal` – median house value in units of $100,000.

Median income is selected as the single numerical input to demonstrate simple linear regression and make the model equation interpretable.

4. Data inspection and preparation The notebook displays the first and last records, dataset dimensions, data types, descriptive statistics, missing-value counts and duplicate-row count. The official dataset documentation reports no missing values. The notebook checks this in the loaded copy. No ID column is used as a predictor. No unsupported row deletion or scaling is applied.

5. Relationship visualisation A scatter plot places `MedInc` on the horizontal axis and `MedHouseVal` on the vertical axis. The plot is used to inspect the association. A visible association does not establish causation.

6. Train-test split The data is divided into 80% training and 20% testing with `random_state=42`. The training set is used to estimate model parameters; the test set is reserved for evaluation on unseen observations.

7. Linear Regression model The fitted model has the form:

`MedHouseVal = intercept + (coefficient × MedInc)`

The actual intercept and coefficient are printed by the executed notebook/script. The coefficient represents the estimated change in target units for a one-unit increase in `MedInc`; the intercept is the model's estimated target when `MedInc` is zero.

8. Predictions and evaluation The notebook/script presents actual and predicted test values and calculates: - MAE – average absolute error. - MSE – average squared

error. - RMSE – error in the target's units. - R^2 – comparison with a mean-based baseline; a negative test R^2 is possible.

Important: Run all notebook cells to populate the actual coefficient, intercept, predictions and metric values. They are intentionally not invented in this report.

9. Actual vs predicted plot The second plot compares actual test values with predictions and includes a 45-degree perfect-prediction reference line. Points closer to the line indicate more accurate predictions.

10. New example prediction The code selects the median `MedInc` value from the training set (within the training feature range) and predicts its median house value. The input and prediction are saved in `outputs/new_prediction.json`. State the resulting value and units after executing the notebook.

11. Conclusion This activity demonstrates a complete introductory regression workflow: dataset inspection, feature and target selection, visualisation, train-test split, Linear Regression training, prediction, metric-based evaluation and a new example prediction. A one-feature model is easy to interpret but cannot represent all factors that affect house prices. The data is historical and aggregated at census-block-group level, so it is not suitable for estimating current prices of individual houses.

Faculty acknowledgement This project was completed as part of the Machine Learning CIE 2 activity. Faculty reference and Machine Learning learning resources: https://github.com/Princy2129